

MD. ESHA RUHULLA



CONTACT

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- Gazipur, Dhaka, Bangladesh

EDUCATION

- Islamic University of Technology**
B.Sc. in Civil & Environmental Engineering
CGPA 3.40 / 4.00
2022 – 2026
- Rangpur Govt. College**
Higher Secondary Certificate (Science)
GPA 5.00 / 5.00
2019 – 2021
- Police Lines School & College, Rangpur**
Secondary School Certificate (Science)
GPA 5.00 / 5.00
2017 – 2019

TECHNICAL SKILLS

- Numerical Modelling:** PLAXIS 3D, PLAXIS 2D, ETABS
- Scientific Computing:** Python (NumPy, SciPy, Pandas, Matplotlib)
- Machine Learning:** DeepXDE, PyTorch (physics-informed neural networks)
- Computer Aided Design:** AutoCAD
- GIS:** ArcGIS
- Documentation:** LaTeX, MS Excel, Word, PowerPoint

EXTRA CURRICULAR ACTIVITIES

- Vice President**—CENNOVATION 2026, Dept. of CEE, IUT
- Head, PR & Communication**—ACI IUT Student Chapter, 2025–2026

CERTIFICATIONS & TRAININGS

- Industrial Training — DOHWA Engineering Co., Ltd. (2025)

REFERENCES

- Dr. Amimul Ahsan**—Associate Professor, CEE, IUT
- Prof. Dr. Hossain Md. Shahin**—Professor & Supervisor, CEE, IUT

CAREER OBJECTIVE

IUT graduate in Civil & Environmental Engineering with experience spanning geotechnical design, three-dimensional finite element modelling & computational analysis, complemented by consultancy training at DOHWA Engineering covering procurement under PPR, quantity surveying, cost estimation & geotechnical investigation practice. Analytical, detail-driven & field-ready, with hands-on PLAXIS 3D foundation design taken to failure & a research background in tunnelling-induced settlement prediction, seeking to build a career in geotechnical design & underground construction with the goal of growing into technical leadership in geotechnical engineering.

INTERNSHIPS

Industrial Trainee — DOHWA Engineering Co., Ltd. *Geotechnical & consultancy practice — Bangladesh Branch, Dhaka*

Oct 2025

- Four-week training across planning, design, project management and procurement workflows at an international consultancy.
- Assisted in preparing EOIs and RFPs and in proposal evaluation under Bangladesh's Public Procurement Rules (PPR); practised quantity surveying and cost estimation.
- Studied geotechnical investigation practice for foundation design; site visit to the Pagla Sewage Treatment Plant, Narayanganj, under the Dhaka Sanitation Improvement Project.

DESIGN & MODELLING PROJECTS

Final Year Design Project (FYDP) *Foundation design of a multi-storeyed building — PLAXIS 3D*

2026

- Built three-dimensional finite element models of a pile foundation and a raft foundation on a common soil profile and loaded each to failure through staged construction.
- Pile foundation: ultimate bearing capacity 1879.85 kN at 21.9 mm maximum total displacement. Raft foundation: 2517.30 kN at 117.8 mm.
- Compared the two systems on capacity against deformation and recommended design parameters on settlement and bearing-capacity criteria.

Analysis Code for Tunnelling Settlement Prediction *Complex-variable solver and PINN implementation — Python*

2025 – 2026

- Reproducible analysis pipeline in NumPy/SciPy and DeepXDE; every measured value read from source data rather than digitised from published figures.

PUBLICATIONS

- A. Mahmud and **M. E. Ruhulla**, “Forward prediction of tunnelling-induced settlement: a unified displacement function and a physics-informed neural network validated against controlled model tests,” *International Conference on Civil Engineering (ICCE 2026)*, Dhaka, 17–19 December 2026. **Submitted, under review.**

UNDERGRADUATE THESIS

Forward Prediction of Tunnelling-Induced Settlement *A unified displacement function and a physics-informed neural network validated against controlled model tests — Supervisor: Prof. Dr. Hossain Md. Shahin*

2025 – 2026

- Tested whether settlement above a shallow tunnel can be predicted forward from the tunnel-wall deformation, rather than by back-fitting a measured trough.
- Implemented the unified displacement function as a complex-variable solution of the elastic half-plane — conformal mapping, Laurent-series potentials and least-squares collocation — with boundary residual 10^{-14} and a published field case reproduced to within 0.02 mm.
- Rebuilt the physics-informed neural network in DeepXDE, matching the analytical benchmark to 0.77% of peak settlement.
- Validated against model tests with imposed tunnel-wall movement: peak settlement within 12% out-of-sample, and mean R^2 raised from 0.41 to 0.91 by an empirical width correction.